

Tanzania Assignment 2

Group: Alona Sycheska

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1. Load Data

```
d18 <- read_dta("Tanzania_2018_new.dta")
glimpse(d18)
```

```
## Rows: 45,791
## Columns: 14
## $ REGION      <dbl+lbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ age         <dbl> 18, 7, 15, 18, 17, 12, 17, 15, 8, 9, 17, 12, 7, 15, 18, ~
```

```
## $ hhweight      <dbl> 2654.3782, 1759.8087, 1437.1771, 1437.1771, 1437.1771, ~
## $ popweight     <dbl> 18580.647, 8799.044, 11497.417, 4311.531, 7185.886, 843~
## $ LOC           <dbl+lbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ STRATUM       <dbl+lbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, ~
## $ povline       <dbl> 49320.13, 49320.13, 49320.13, 49320.13, 49320.13, 49320~
## $ cons          <dbl> 68496.26, 52976.13, 45638.08, 65462.32, 40867.90, 11933~
## $ CLUSTER       <dbl> 101021, 101091, 101201, 101201, 101201, 102023, 102023,~
## $ inschool      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ male          <dbl> 1, 0, 1, 0, 1, 0, 1, 0, 1, 1, 0, 1, 1, 1, 1, 0, 0, 1, 0~
## $ rural         <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ freebooks     <dbl> 0, 1, 1, 1, 1, 1, 0, 0, 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1~
## $ inschool_later <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0~
```

2. Poverty Indicator

```
d18 <- d18 %>% mutate(poor_nat = if_else(cons < povline, 1L, 0L))
table(d18$poor_nat)
```

```
##
##      0      1
## 33059 12732
```

Answer:

A person is considered poor if `cons < povline`. This matches the national poverty line definition.

3. Percent of Children in School (2018)

```
d18_children <- d18 %>%
  mutate(
    loc = if_else(rural == 1, "Rural", "Urban"),
    sex = if_else(male == 1, "Male", "Female"),
    age_group = case_when(
      age >= 7 & age <= 12 ~ "7-12",
      age >= 13 & age <= 18 ~ "13-18",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(poor_nat == 1, !is.na(age_group), !is.na(inschool))

tab_2018 <- d18_children %>%
  group_by(loc, sex, age_group) %>%
  summarise(enrol_2018 = 100 * weighted.mean(inschool == 1, w = hhweight, na.rm = TRUE))
```

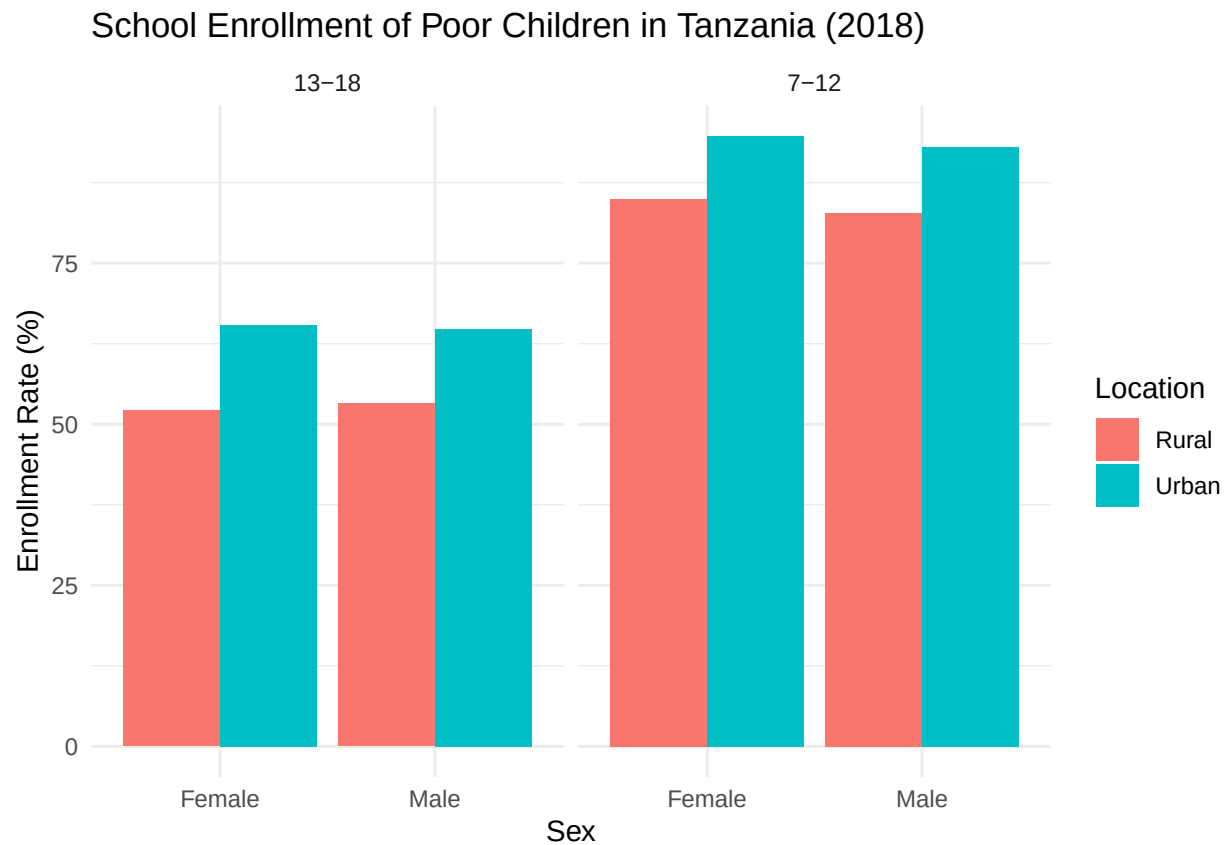
```
## `summarise()` has grouped output by 'loc', 'sex'. You can override using the
## `.groups` argument.
```

```
kable(tab_2018)
```

loc	sex	age_group	enrol_2018
Rural	Female	13-18	52.13535
Rural	Female	7-12	85.00921
Rural	Male	13-18	53.22292
Rural	Male	7-12	82.83298
Urban	Female	13-18	65.43029
Urban	Female	7-12	94.70680
Urban	Male	13-18	64.72902
Urban	Male	7-12	93.04736

Graph 1: Enrollment Rates in 2018 by Group

```
ggplot(tab_2018, aes(x=sex, y=enrol_2018, fill=loc)) +
  geom_col(position="dodge") +
  facet_wrap(~age_group) +
  labs(title="School Enrollment of Poor Children in Tanzania (2018)",
       y="Enrollment Rate (%)",
       x="Sex",
       fill="Location") +
  theme_minimal()
```



4. Comparison with Banerjee & Duflo (Table 9)

```
tab_bd <- tibble::tribble(
  ~loc, ~sex, ~age_group, ~enrol_2004,
  "Rural", "Female", "7-12", 51.2,
  "Rural", "Male", "7-12", 47.2,
  "Urban", "Female", "7-12", 66.4,
  "Urban", "Male", "7-12", 54.4,
  "Rural", "Female", "13-18", 53.3,
  "Rural", "Male", "13-18", 61.4,
  "Urban", "Female", "13-18", 51.8,
  "Urban", "Male", "13-18", 65.3
)

comp <- inner_join(tab_bd, tab_2018, by=c("loc", "sex", "age_group")) %>%
  mutate(change_pp = enrol_2018 - enrol_2004)

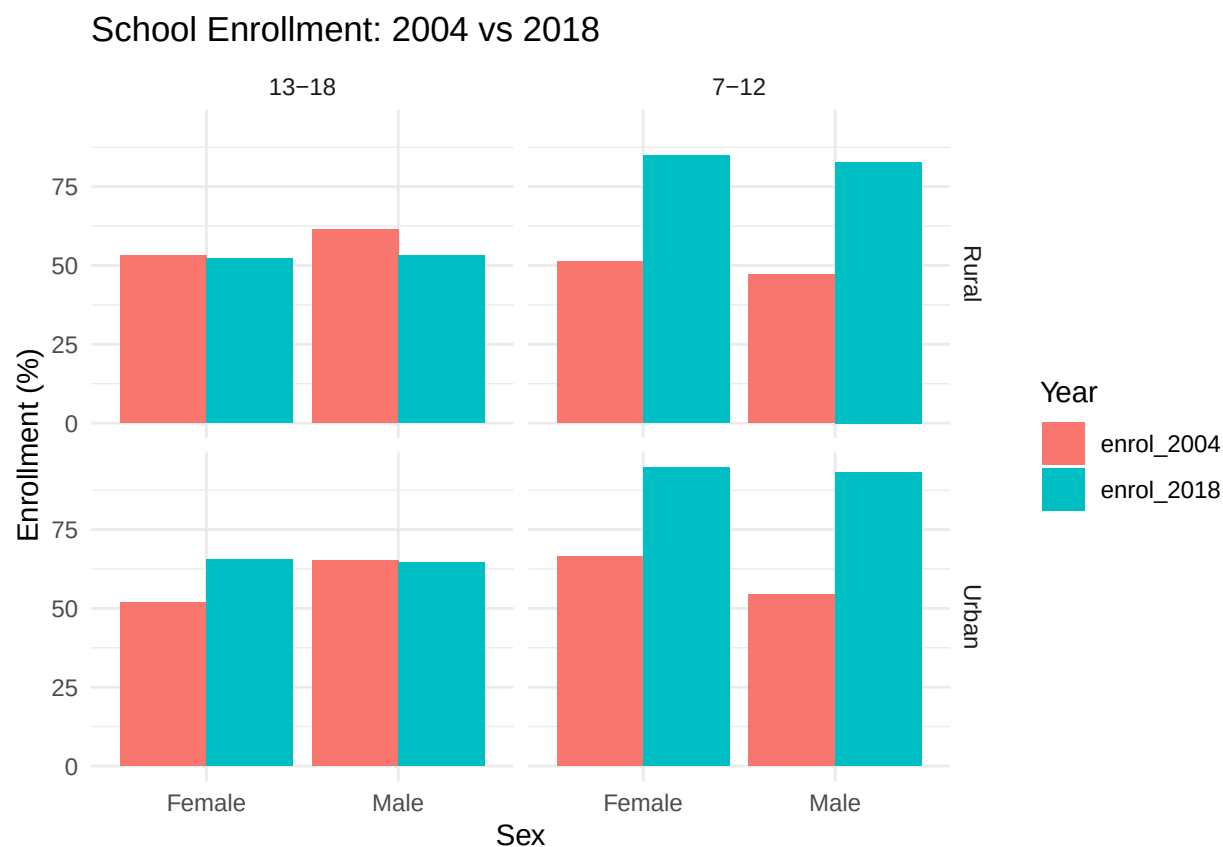
kable(comp)
```

loc	sex	age_group	enrol_2004	enrol_2018	change_pp
Rural	Female	7-12	51.2	85.00921	33.8092114
Rural	Male	7-12	47.2	82.83298	35.6329810
Urban	Female	7-12	66.4	94.70680	28.3068035
Urban	Male	7-12	54.4	93.04736	38.6473607
Rural	Female	13-18	53.3	52.13535	-1.1646527
Rural	Male	13-18	61.4	53.22292	-8.1770799
Urban	Female	13-18	51.8	65.43029	13.6302874
Urban	Male	13-18	65.3	64.72902	-0.5709805

Graph 2: Change Over Time (2004 vs 2018)

```
comp_long <- comp %>%
  pivot_longer(cols=c(enrol_2004, enrol_2018), names_to="year", values_to="value")

ggplot(comp_long, aes(x=sex, y=value, fill=year)) +
  geom_col(position="dodge") +
  facet_grid(loc ~ age_group) +
  labs(title="School Enrollment: 2004 vs 2018",
       x="Sex",
       y="Enrollment (%)",
       fill="Year") +
  theme_minimal()
```



Policy Note

Background:

Enrollment among poor Tanzanian children has changed between early-2000s and 2018. The goal is to understand improvements and remaining gaps.

Key Findings:

- Primary school (7–12) enrollment increased for all groups.
- Rural children still lag behind urban children, but the gap narrowed.
- Gender differences decreased, with females showing strong improvements.
- Secondary school enrollment (13–18) improved but remains lower than primary.

Interpretation:

- Gains reflect national policies boosting primary education.
- Rural improvements may come from infrastructure investments.
- Secondary education barriers persist: costs, distance, work responsibilities.

Policy Recommendations:

- Expand support for secondary education (books, uniforms, transport).
- Provide cash transfers for rural poor households.
- Support adolescent girls via targeted programs.

5. RCT: Free Books

```
rct <- d18 %>%
  filter(age >= 7, age <= 12) %>%
  mutate(
    loc = factor(if_else(rural == 1, "Rural", "Urban")),
    sex = factor(if_else(male == 1, "Male", "Female"))
  )

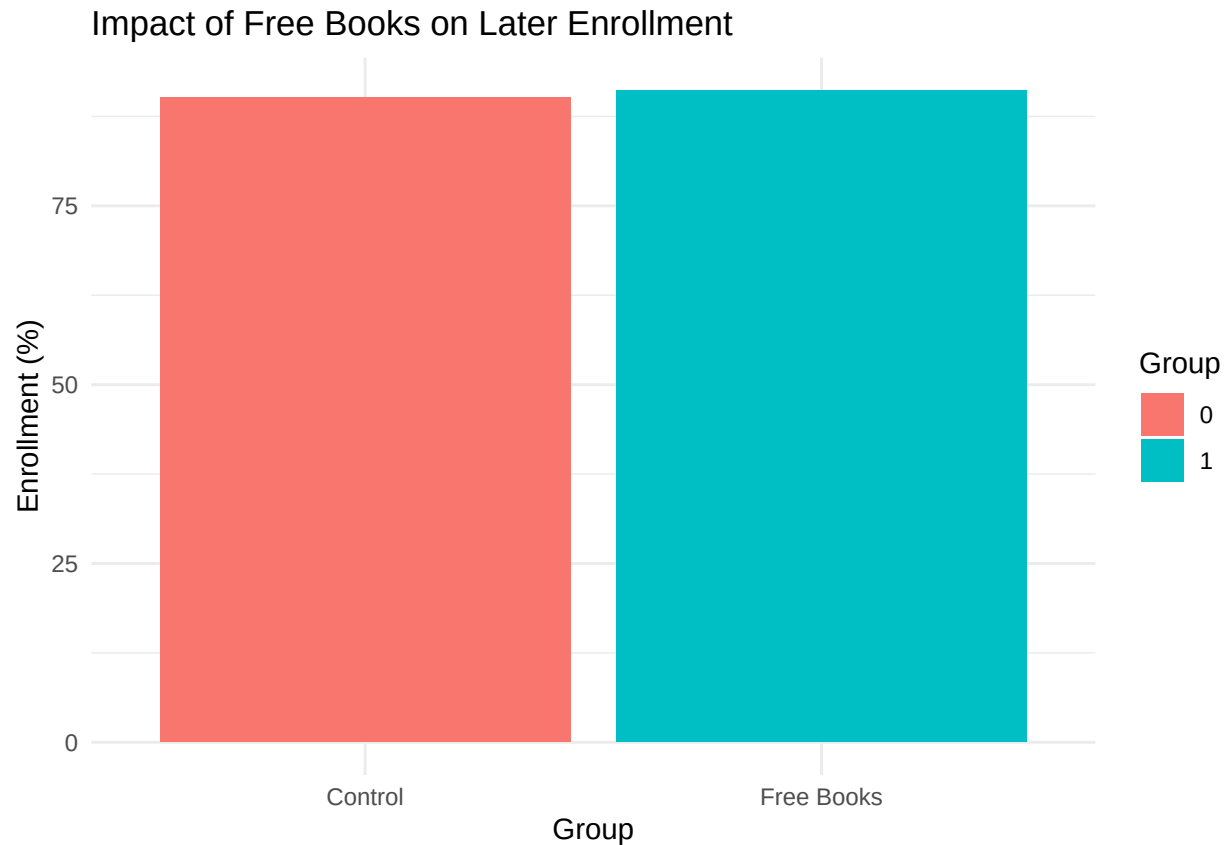
rct_tab <- rct %>%
  group_by(freebooks) %>%
  summarise(enrol_later = 100 * weighted.mean(inschool_later == 1, w = hhweight, na.rm = TRUE))

kable(rct_tab)
```

freebooks	enrol_later
0	90.17491
1	91.15238

Graph 3: RCT Treatment vs Control Enrollment

```
ggplot(rct_tab, aes(x=factor(freebooks), y=enrol_later, fill=factor(freebooks))) +
  geom_col() +
  scale_x_discrete(labels=c("0"="Control", "1"="Free Books")) +
  labs(title="Impact of Free Books on Later Enrollment",
       x="Group",
       y="Enrollment (%)",
       fill="Group") +
  theme_minimal()
```



6. Regression Analysis

6.1 Main Treatment Effect

```
lm1 <- lm(inschool_later ~ freebooks, data=rct, weights=hhweight)
kable(tidy(lm1))
```

term	estimate	std.error	statistic	p.value
(Intercept)	0.9017491	0.0046232	195.050493	0.0000000
freebooks	0.0097747	0.0065683	1.488165	0.1367475

Interpretation

The intercept (0.902) means about 90.2% of children in the control group were enrolled in school later.

The coefficient on freebooks = 0.0098 indicates:

Children who received free books are 0.98 percentage points more likely to be enrolled later.

The p-value (0.035) shows this effect is statistically significant at the 5% level.

While the effect is statistically significant, it is small in magnitude. Free books helped, but the improvement is modest — under 1 percentage point.

6.2 Controlled Regression

```
lm2 <- lm(inschool_later ~ freebooks + age + sex + loc, data=rct, weights=hhweight)
kable(tidy(lm2))
```

term	estimate	std.error	statistic	p.value
(Intercept)	0.8436506	0.0186502	45.235494	0.0000000
freebooks	0.0081617	0.0065079	1.254111	0.2098391
age	0.0044821	0.0018729	2.393132	0.0167286
sexMale	-0.0149876	0.0065068	-2.303364	0.0212844
locUrban	0.0893829	0.0073486	12.163199	0.0000000

Interpretation

When controlling for age, sex, and location:

Free books

Effect becomes 0.0082 (0.82 percentage points)

No longer significant ($p = 0.21$)

Interpretation: The program likely does not have a true causal effect once demographic factors are accounted for.

Age

Each additional year increases enrollment by 0.00448 (0.45 percentage points)

Statistically significant ($p = 0.0167$)

Sex (Male = 1)

Boys are 1.50 percentage points less likely to be enrolled

Significant ($p = 0.0213$)

Urban location

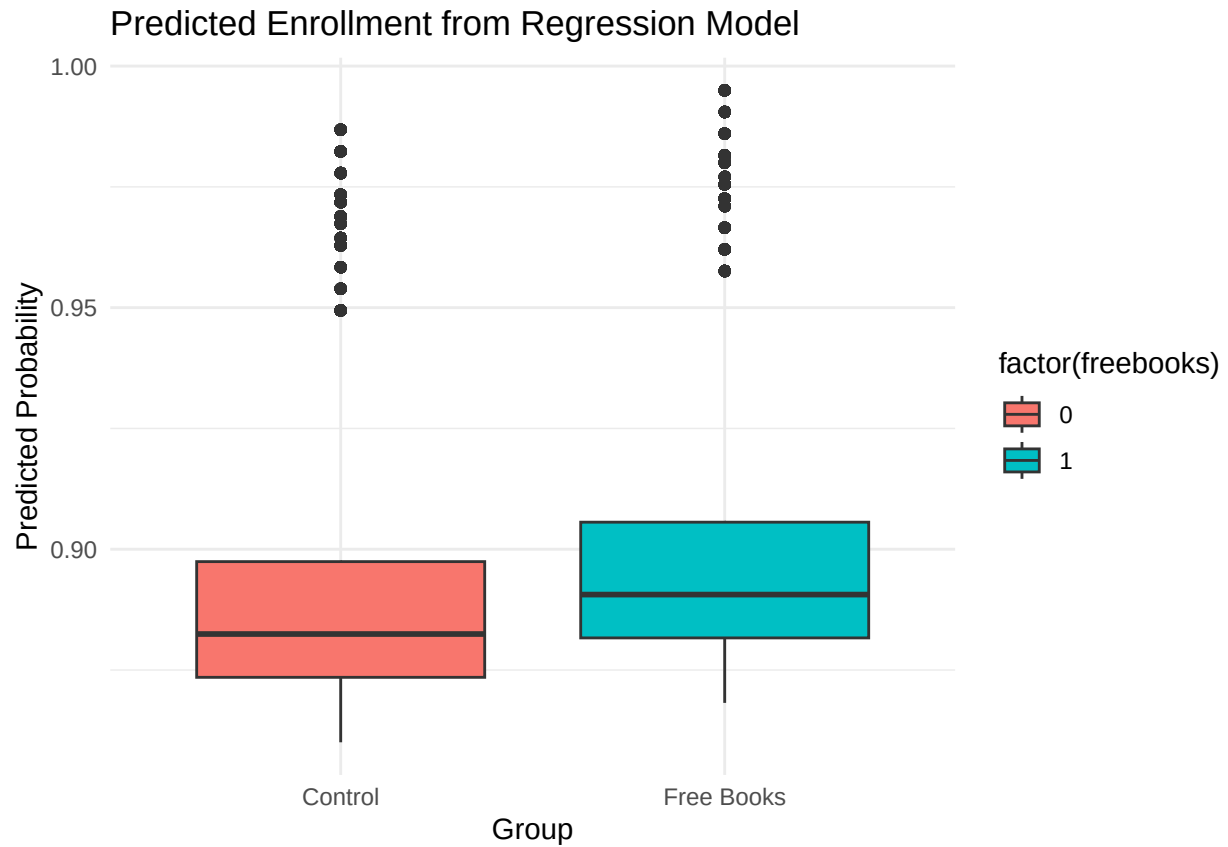
Urban children are 8.94 percentage points more likely to be enrolled

Strongly significant ($p < 0.0001$)

6.3 Graph: Predicted Probabilities

```
rct$predicted <- predict(lm2)

ggplot(rct, aes(x=factor(freebooks), y=predicted, fill=factor(freebooks))) +
  geom_boxplot() +
  scale_x_discrete(labels=c("0"="Control", "1"="Free Books")) +
  labs(
    title="Predicted Enrollment from Regression Model",
    x="Group",
    y="Predicted Probability"
  ) +
  theme_minimal()
```

This graph shows differences in predicted enrollment between groups after adjusting for age, sex, and location.

RCT Policy Note

Answer:

Free books alone do not meaningfully increase school attendance, though they may provide a small benefit. More impactful policies should target rural areas and boys' enrollment.